Diego F. Rojas-Torres

Biomedical Engineer · PhD Candidate in Engineering and Sciences with the Industry · Health Systems Researcher · Digital Health Leader

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RESEARCH PROFILE

Biomedical engineer and PhD candidate working at the intersection of oncology, healthcare operations, health data science, and digital transformation. Research interests include oncology care pathways, dynamic treatment prioritization, process mining, discrete-event simulation, waiting-list management, and data-driven decision support under resource constraints.

CURRENT APPOINTMENTS

Visiting Researcher — Department of Industrial and Operations Engineering, University of Michigan, Ann Arbor, USA
August 3, 2026 – December 18, 2026; working with Professor Mariel S. Lavieri.

Researcher and Head of Digital Transformation — National Cancer Institute of Chile, Santiago, Chile
2023 – present

PhD Candidate — Engineering and Sciences with the Industry, Pontificia Universidad Católica de Chile
2021 – present

EDUCATION

PhD (candidate), Engineering and Sciences with the Industry — Pontificia Universidad Católica de Chile, 2021–present.
Postgraduate Diploma in Healthcare Institution Management — Universidad de Santiago de Chile, 2018–2019.
Civil Biomedical Engineer — Universidad de Valparaíso, 2010–2016.

DOCTORAL RESEARCH

Simulation-Based Patient Management and Data- and Process-Mining-Driven Waiting-List Administration: An Applied Oncology Case Study.
Original Spanish title: *Modelo de gestión de pacientes mediante simulación y administración de listas de espera basada en minería de datos y procesos: caso aplicado del paciente oncológico.*

RESEARCH OUTPUTS

Under peer review — Rojas-Torres, D., Mac Cawley, A., Lavieri, M. S., Ruiz, A., Abarca, I., & Flores, F. *Prioritizing Access to Cancer Treatment in Low-Resource Countries: A Systematic Review.* Public Health Reviews. Under peer review after passing initial editorial screening. PROSPERO: CRD420261334566.

Manuscript in preparation — *A Conceptual Framework for a Prognostic Decision-Support Tool in Oncology Treatment Prioritization under Resource Constraints: Case Illustrations from a Tertiary Cancer Center.*

Manuscript in preparation — *Redesigning Cancer Patient Care by Linking Oncology Care Pathways, Resource Capacity, and Patient Outcomes: A Process Mining and Simulation Case Study at a High-Complexity Cancer Center.*

Peer-reviewed publication — González, X., Cerda, B., Ciudad, A. M., Abarca, I., Rojas-Torres, D., et al. (2025). Cancer research in Chile: beyond universities, contributions from the National Cancer Institute. *International Journal of Medical and Surgical Sciences, 12*(1), 1–4. DOI: 10.32457/ijmss.v12i1.2997.

Peer-reviewed publication — Mansilla, E., Mazzon, E., Cárcamo, D., Jurado, F., Lara, L., Rojas-Torres, D., et al. (2019). Telestroke in Chile: 1 year experience at 7 hospitals. *Revista Médica de Chile, 147*(9), 1107–1113. DOI: 10.4067/S0034-98872019000901107.

Peer-reviewed publication — Mazzon, E., Delfino, C., Mirelis, S., Rojas-Torres, D., et al. (2022). Breaking Down Barriers: Easter Island's First Telestroke Thrombolysis Experience and Case Report. *Journal of Stroke and Cerebrovascular Diseases, 31*(6), 106412. DOI: 10.1016/j.jstrokecerebrovasdis.2022.106412.

PRESENTATIONS

2026 — Transparent Data-Driven Oncology Prioritization: A Pathway-Specific Proof of Concept. Poster, 2026 SIAM Student Mini-Symposium in Applied Mathematics, University of Michigan, August 29, 2026. Sponsored by MCAIM.

2025 — Data-Driven Cancer Patient Waiting List Management. HOPER 2025 School on Health Operations Research, Université du Québec à Montréal. IFORS–ALIO Travel Scholarship recipient.

2025 — Advanced Management of Oncology Waiting Lists Using Prospective Methods of Prioritization. Biomedical Engineering Colloquium Series, University of Valparaíso.

PROFESSIONAL EXPERIENCE

Digital health and healthcare systems: National Cancer Institute of Chile; Servicio de Salud Metropolitano Sur; Plataforma 360 Ltda.; Hospital Clínico Universidad de Chile. Experience in digital transformation, health information systems, telemedicine, clinical technology, process improvement, and coordination of clinical and technical teams.

Selected projects: National TeleStroke Network in Chile; TeleTriage primary-care demand management; telemedicine for older adults; rehabilitation and telerehabilitation research; digital and electronic systems for delirium prevention in hospitalized patients.

AWARDS, TEACHING & METHODS

Awards: ANID National Doctoral Scholarship (2022); IFORS–ALIO Travel Scholarship for HOPER 2025.

Teaching: Guest Lecturer, Medical Informatics, Universidad del Desarrollo (2023–2024); Collaborative Learning Program Mentor, Universidad de Valparaíso (2015).

Methods and tools: Process mining; discrete-event simulation; optimization models; machine learning; health data science; survival analysis; Python; R; SQL; MATLAB; HL7/FHIR interoperability; C++; clinical and biomedical technology.

ACADEMIC PROFILES

ORCID: https://orcid.org/0009-0008-2136-9110
Google Scholar: https://scholar.google.com/citations?user=VbLf-CYAAAAJ&hl;=es
ResearchGate: https://www.researchgate.net/profile/Diego-Rojas-54
GitHub: https://github.com/diegofrt